

Beyond Nash Equilibrium: Tholonic Triadic Dynamics as a Sustainability-Oriented Alternative to Classical Game Theory in Supply Chain and Value Chain Systems

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Abstract

Classical game theory, grounded in Nash equilibrium and rational-actor assumptions, has dominated the formal analysis of strategic interaction in economics, supply chain management, and mechanism design for over seven decades. Yet its core predictions—particularly the tragedy of the commons and the prisoner’s dilemma—remain descriptive of failure rather than prescriptive of sustainability. This paper introduces the *tholonic model*, a triadic framework in which every system is decomposed into three archetypal forces: *Definition* ($\mathcal{D}$, constraint/structure), *Contribution* ($\mathcal{C}$, integration/flow), and *Negotiation* ($\mathcal{N}$, emergent equilibrium). We show formally that the tholonic sustainability condition— $\mathcal{D} \approx \mathcal{C}$, minimizing maintenance energy—generalises the Nash equilibrium concept while embedding resource-awareness and hierarchical self-similarity absent from standard game-theoretic models. Through analytical comparison, agent-based simulation, and application to multi-phase supply chain and value chain networks, we demonstrate that tholonic dynamics produce cooperative equilibria that resist commons collapse, predict phase-specific bottlenecks invisible to aggregate game-theoretic analysis, and provide a unified optimisation framework spanning physical logistics, economic value creation, and sys-

temic resilience. We discuss implications for economics, operations research, systems theory, and sustainability science.

Keywords: game theory, tholonic model, N-D-C dynamics, supply chain optimization, value chain analysis, sustainability, complex adaptive systems, cooperative equilibrium, tragedy of the commons, systems theory

1 Introduction

The analytical toolkit available to economists, operations researchers, and supply chain practitioners for modelling strategic interaction has been overwhelmingly shaped by non-cooperative game theory [von Neumann and Morgenstern, 1944, Nash, 1950, 1951]. The central solution concept—Nash equilibrium—identifies states from which no individual agent can unilaterally improve its payoff. While mathematically elegant, Nash equilibrium is agnostic to the sustainability of the state it describes: a tragedy-of-the-commons outcome, in which a shared resource is depleted to the detriment of all participants, is as valid a Nash equilibrium as a Pareto-optimal allocation [Hardin, 1968, Ostrom, 1990].

This limitation is not merely academic. In commodity supply chains—gold, agricultural products, rare earths—the physical system imposes constraints that classical game theory

treats as exogenous parameters rather than endogenous dynamics. A gold refinery operating at 99.99% purity standards (a *definitional* constraint) must simultaneously maintain flexible supplier networks (a *contributive* integration); the emergent operational state is neither purely competitive nor purely cooperative, but a dynamic negotiation between structure and flow. Standard game-theoretic models flatten this triadic structure into a payoff matrix, losing the information that determines long-term viability.

This paper presents the **tholonic model** as an alternative and extension. The tholonic framework decomposes every system—from an individual transaction to an entire industry—into three interacting forces:

- $\mathcal{D}$ (**Definition**): constraint, boundary, specification, internal coherence;
- $\mathcal{C}$ (**Contribution**): integration, connection, flow, external exchange;
- $\mathcal{N}$ (**Negotiation**): the emergent equilibrium that arises from $\mathcal{D}$ – $\mathcal{C}$ interaction.

The tholonic model is not a repudiation of game theory; it is a *re-grounding* of strategic interaction within a physically and thermodynamically informed ontology. Where game theory asks “What is the rational strategy?”, the tholonic model asks “What configuration is sustainable?”—and it provides a formal criterion: systems minimise maintenance energy when $\mathcal{D} \approx \mathcal{C}$, a condition we term *tholonic balance*.

The remainder of this paper is organised as follows. Section 2 situates the tholonic model within systems theory and philosophy of science. Section 3 reviews classical game theory and identifies its structural limitations. Section 4 formalises the tholonic framework mathematically. Section 5 provides a rigorous side-by-side comparison. Section 6 applies both frameworks to multi-phase supply chain modelling. Section 7 extends the analysis to value chains and economic surplus. Section 8 presents agent-based simulation results. Section 10 discusses implications and limitations. Section 11 concludes.

2 Philosophical and Systems-Theoretic Foundations

2.1 From Atomism to Triadic Ontology

Classical game theory inherits the atomistic ontology of neoclassical economics: agents are fundamental, preferences are given, and interaction is mediated through payoff functions defined over discrete strategy sets [Mas-Colell et al., 1995]. This framework is profoundly *dualistic*: it recognises only two categories—agents and their environment—and models interaction as bilateral exchange.

The tholonic model draws instead from a *triadic* ontological tradition with deep roots in process philosophy [Whitehead, 1929], general systems theory [von Bertalanffy, 1968], and dissipative structures [Prigogine and Stengers, 1984]. In the Peircean semiotic tradition, every sign relation requires three irreducible components: sign, object, and interpretant [Peirce, 1931]. Analogously, every tholonic system requires three irreducible forces: structure ($\mathcal{D}$), flow ($\mathcal{C}$), and the emergent state ($\mathcal{N}$) that mediates between them.

This triadic structure is not an arbitrary choice. It reflects a fundamental observation about complex systems: *no system can be understood through its constraints alone or its connections alone*; it is the interaction between boundary-making and boundary-crossing that generates observable reality. This insight is shared by autopoietic theory [Maturana and Varela, 1980], where systems maintain identity (analogous to $\mathcal{D}$) through interaction with their environment (analogous to $\mathcal{C}$), producing self-referential organisation (analogous to $\mathcal{N}$).

2.2 Thermodynamic Grounding

The tholonic model embeds thermodynamic principles that game theory lacks. In classical game theory, the “cost” of a strategy is defined purely in terms of opportunity cost within the payoff structure; there is no concept of the *energy required to maintain a strategic position*.

The tholonic framework introduces *maintenance energy*:

$$E_{\text{maint}} = |\mathcal{D} - \mathcal{C}|^\alpha + E_{\text{base}} \quad (1)$$

where $\alpha \geq 2$ is a system-specific exponent and $E_{\text{base}} > 0$ is the irreducible baseline energy cost of existence. This formulation has direct analogues in Landauer’s principle [Landauer, 1961]—the minimum energy cost of information processing—and in the thermodynamics of self-organisation [England, 2013], where dissipative structures must continuously expend energy to maintain themselves far from equilibrium.

Sustainability is then defined as the inverse of maintenance energy:

$$S(\mathcal{D}, \mathcal{C}) = \frac{1}{|\mathcal{D} - \mathcal{C}|^\alpha + E_{\text{base}}} \quad (2)$$

This provides a principled, physically grounded objective function that is entirely absent from classical game theory.

2.3 Self-Similarity and Hierarchical Composition

A distinguishing feature of the tholonic model is its *fractal architecture*. Each tholon ($\mathfrak{T}$) at scale i decomposes into a complete $\mathcal{N}$ - $\mathcal{D}$ - $\mathcal{C}$ structure at scale $i - 1$:

$$\mathfrak{T}_i = \mathcal{N}_i(\mathcal{D}_i(\mathfrak{T}_{i-1}^{(D)}), \mathcal{C}_i(\mathfrak{T}_{i-1}^{(C)})) \quad (3)$$

This recursive composition is isomorphic to the renormalisation group in statistical physics [Wilson, 1971], where macroscopic behaviour emerges from iterated application of the same dynamical laws across scales.

Game theory possesses no native mechanism for hierarchical self-similarity. While extensive-form games can represent sequential decisions and subgame perfection [Selten, 1965] provides refinement within hierarchies, the underlying strategic structure does not repeat across scales—a payoff matrix at the firm level bears no structural resemblance to a payoff matrix at the industry level. The tholonic model’s self-similarity, by contrast, ensures that insights gained at any scale transfer to all others, a property of enormous practi-

cal value in supply chain analysis where micro-transactions, meso-relationships, and macro-networks must be analysed coherently.

3 Classical Game Theory: Foundations and Limitations

3.1 Nash Equilibrium and Its Discontents

A *normal-form game* $\Gamma = (I, \{S_i\}_{i \in I}, \{u_i\}_{i \in I})$ consists of a finite set of players I , strategy sets S_i , and payoff functions $u_i : \prod_j S_j \rightarrow \mathbb{R}$. A strategy profile $s^* = (s_1^*, \dots, s_n^*)$ is a *Nash equilibrium* if:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall i \in I, \forall s_i \in S_i \quad (4)$$

The Nash equilibrium concept is subject to well-documented limitations:

- (i) **Multiplicity:** Games typically admit multiple Nash equilibria with no principled selection criterion [Harsanyi and Selten, 1988].
- (ii) **Sustainability blindness:** The equilibrium concept is *static*; it identifies fixed points but says nothing about whether they can be maintained over time under resource constraints.
- (iii) **Commons collapse:** In public goods games, the unique Nash equilibrium is universal defection—the tragedy of the commons [Hardin, 1968]—despite its collective irrationality.
- (iv) **Exogenous structure:** The game’s structure (players, strategies, payoffs) is given *a priori*; there is no mechanism for the game itself to evolve.
- (v) **No resource dynamics:** Payoff functions are fixed and do not depend on accumulated resource levels or system health.

3.2 Extensions and Their Insufficiency

Mechanism design [Myerson, 1981] and evolutionary game theory [Maynard Smith, 1982]

partially address these limitations. Mechanism design constructs incentive-compatible institutions but requires a principal with complete information about the game’s structure—a condition rarely met in supply chains. Evolutionary game theory introduces population dynamics and selection but operates on *strategies*, not on the *structural configuration* of the system itself.

Repeated games with discounting [Fudenberg and Tirole, 1991] can sustain cooperation through folk-theorem arguments, but the resulting equilibria are sustained by *threat* (reversion to punishment strategies), not by *structural balance*. The cooperation is fragile: it depends on precise beliefs about discount factors and monitoring capacity.

Ostrom’s institutional analysis [Ostrom, 1990, 2005] demonstrates empirically that communities develop governance structures to avoid commons tragedy, but this work lacks a formal mathematical framework connecting institutional design to system sustainability.

The tholonic model, as we show below, provides such a framework.

4 The Tholonic Framework: Mathematical Formalisation

4.1 Definitions

Definition 4.1 (Tholon). *A tholon is a triple $\mathfrak{T} = (\mathcal{D}, \mathcal{C}, \mathcal{N})$ where:*

- $\mathcal{D} \in \mathbb{R}_+^n$ is the Definition vector (constraint intensities);
- $\mathcal{C} \in \mathbb{R}_+^m$ is the Contribution vector (integration intensities);
- $\mathcal{N} : \mathbb{R}_+^n \times \mathbb{R}_+^m \rightarrow \mathbb{R}_+$ is the Negotiation function producing the emergent state.

Definition 4.2 (Aggregate Magnitudes). *Given weight vectors $\mathbf{w}^D \in \mathbb{R}_+^n$ and $\mathbf{w}^C \in \mathbb{R}_+^m$:*

$$D := \sum_{i=1}^n w_i^D \cdot d_i = \mathbf{w}^D \cdot \mathcal{D} \quad (5)$$

$$C := \sum_{j=1}^m w_j^C \cdot c_j = \mathbf{w}^C \cdot \mathcal{C} \quad (6)$$

4.2 The Negotiation Operator

The central equation of the tholonic model defines the emergent $\mathcal{N}$ -state as the energy-minimising configuration:

$$\mathcal{N}^* = \arg \min_{\mathcal{N}} [E_D + E_C + E_{\text{int}}(\mathcal{D}, \mathcal{C}, \mathcal{N})] \quad (7)$$

Under the specific functional form used in simulation:

$$N = \sqrt{D \cdot C} \cdot \beta(D, C) \quad (8)$$

where the *balance factor* β is:

$$\beta(D, C) = \exp\left(-\frac{2|D - C|}{\max(D, C)}\right) \quad (9)$$

Proposition 4.1 (Balance Maximises N). *For fixed $D + C = K$ (constant total resources), N is maximised when $D = C = K/2$.*

Proof. With $D + C = K$ and $D = K/2 + \delta$, $C = K/2 - \delta$:

$$N(\delta) = \sqrt{\frac{K^2}{4} - \delta^2} \exp\left(-\frac{4|\delta|}{K/2 + |\delta|}\right)$$

The geometric mean $\sqrt{(K/2)^2 - \delta^2}$ is maximised at $\delta = 0$ (AM-GM inequality), and β is maximised at $\delta = 0$ (exponent is zero). Since both factors are maximised at $\delta = 0$, their product is maximised at $D = C = K/2$. $\square$

4.3 Sustainability and Energy Landscape

Definition 4.3 (Tholonic Sustainability). *The sustainability of a tholon is:*

$$S(D, C) = \frac{100}{|D - C|^2 + E_{\text{base}}} \quad (10)$$

where $E_{\text{base}} > 0$ is the baseline energy cost.

The energy landscape over (D, C) -space defines a potential surface:

$$U(D, C) = (D - C)^2 + E_{\text{base}} + \lambda_D \sum_i \phi_D(d_i) + \lambda_C \sum_j \phi_C(c_j) \quad (11)$$

where ϕ_D and ϕ_C are per-parameter cost functions (quadratic for $\mathcal{D}$, superlinear for $\mathcal{C}$ due to coordination overhead).

4.4 Stability Analysis

The dynamic evolution of a tholon is governed by:

$$\frac{d}{dt} \begin{pmatrix} D \\ C \\ N \end{pmatrix} = \begin{pmatrix} \alpha_D(D^* - D) - \beta_D \cdot D \cdot P \\ \alpha_C(C^* - C) - \beta_C \cdot C \cdot P \\ k_1 DC - k_2 N - k_3 N |D - C| \end{pmatrix} \quad (12)$$

where D^*, C^* are targets, P is cost pressure, and k_1, k_2, k_3 are rate constants.

Proposition 4.2 (Stability at Balance). *The equilibrium point $D^* = C^*$ is a stable node: all eigenvalues of the Jacobian $\mathbf{J}$ evaluated at (D^*, C^*, N^*) with $D^* = C^*$ have strictly negative real parts.*

Proof sketch. At $D = C$, the $|D - C|$ terms vanish and the cross-coupling terms in the Jacobian reduce to symmetric form. The diagonal elements $\partial F_D / \partial D = -\alpha_D - \beta_D P$ and $\partial F_C / \partial C = -\alpha_C - \beta_C P$ are strictly negative for positive parameters. The off-diagonal coupling through N is bounded, and the Routh–Hurwitz conditions are satisfied for physically meaningful parameter ranges. $\square$

5 Formal Comparison: Game Theory vs. Tholonic Dynamics

5.1 Structural Correspondence

We establish a mapping between the two frameworks:

Table 1: Structural correspondence between game-theoretic and tholonic concepts.

Concept	Game Theory	Tholonic Model
Agents	Players I	Tholons $\mathfrak{T}$
Choices	Strategy set S_i	$\mathcal{D}$ – $\mathcal{C}$ config.
Outcome	Payoff $u_i(s)$	$\mathcal{N}$ -state $N(D, C)$
Equilib.	Nash: no unilateral deviation gain	Balance: $D \approx C$ minimises energy
Optimal.	Pareto efficiency	Sustainability S
Dynamics	Replicator eq.	Equation (12)
Hierarchy	Subgames	Thologram recursion
Resource	Exogenous	Endogenous $R(t)$

5.2 The Public Goods Game: A Comparative Analysis

Consider n agents sharing a common resource $R \in [0, 1]$. Each round, agent i chooses to cooperate ($a_i = 1$, cost c) or defect ($a_i = 0$). The public good payoff is $g = b \cdot R \cdot \sum_i a_i / n$, where b is the benefit multiplier.

5.2.1 Standard Game-Theoretic Analysis

Each agent updates cooperation probability p_i via reinforcement:

$$p_i \leftarrow p_i + \eta \cdot \begin{cases} +1 & \text{if } a_i = 1 \text{ and } \pi_i > 0 \\ -1 & \text{if } a_i = 1 \text{ and } \pi_i \leq 0 \end{cases} \quad (13)$$

where $\pi_i = g - a_i \cdot c$ and η is the learning rate. The resource evolves as:

$$R_{t+1} = \text{clamp}(R_t + r_{\text{reg}} \cdot \bar{a} - r_{\text{dec}} \cdot (1 - \bar{a})) \quad (14)$$

where $\bar{a} = \sum_i a_i / n$ is the cooperation fraction.

Under standard analysis, when $c > b \cdot R / n$ (which holds for moderate n), the individually rational strategy is defection. The unique Nash equilibrium is $p_i \rightarrow 0 \forall i$, leading to $R \rightarrow 0$: commons collapse.

5.2.2 Tholonic Analysis

The tholonic model augments each agent’s update with a *sustainability signal*:

$$\pi_i^{\text{tho}} = \pi_i + \lambda \cdot (1 - R) \cdot (2a_i - 1) \quad (15)$$

where $\lambda \geq 0$ is the tholonic coupling parameter. When R is low (resource threatened), cooperators receive a bonus $+\lambda(1 - R)$ and defectors receive a penalty $-\lambda(1 - R)$. This corresponds to each agent receiving information about the $\mathcal{D}$ – $\mathcal{C}$ balance of the parent system:

- Low R indicates $\mathcal{D}$ -dominance (resource depletion = over-constraint);

- The λ signal encodes the tholonic principle that contribution to the parent context is essential for the child’s viability.

Theorem 5.1 (Tholonic Cooperation Sustainability). *For $\lambda > c/(1 - R_{\min})$ where $R_{\min}$ is the minimum stable resource level, the tholonic public goods game admits a stable cooperative equilibrium with $p_i > 0.5$ for all i and $R > R_{\min}$.*

Proof sketch. At the cooperative equilibrium, $\sum a_i/n \approx \bar{p}$ and R stabilises at $R^* = r_{\text{regen}}\bar{p}/(r_{\text{regen}}\bar{p} + r_{\text{decay}}(1 - \bar{p}))$. The tholonic payoff for a cooperator is $\pi_{\text{coop}} = g - c + \lambda(1 - R^*)$. For this to be positive (sustaining cooperation), we need $\lambda(1 - R^*) > c - g$, which holds when λ exceeds $c/(1 - R_{\min})$. Stability follows from the contraction mapping property: higher cooperation $\rightarrow$ higher $R \rightarrow$ lower sustainability signal $\rightarrow$ equilibrium (negative feedback). $\square$

5.3 Comparative Properties

Table 2: Comparative properties in the public goods setting.

Property	Game Theory	Tholonic Model
Equilibrium	Universal defec- tion	Cooperative for $\lambda > \lambda^*$
Resource	Collapse ($R \rightarrow 0$)	Sustained ($R > R_{\min}$)
Information	Payoff only	Payoff + system state
Cooperation Scalability	Threat-based Fixed player set	Structural balance Hierarchical recur- sion
Resolution Objective	Aggregate Individual payoff	Per-phase $\mathcal{D}\text{-}\mathcal{C}$ System sustainabil- ity

The crucial distinction is *ontological*: game theory treats the resource as a passive back-drop; the tholonic model treats the resource as a constitutive element of the system whose health feeds back into agent decision-making.

6 Application to Supply Chain Economics

6.1 Supply Chains as Tholonic Structures

A supply chain is a sequence of transformations that convert raw inputs into finished goods, each characterised by physical state changes, custody transfers, and measurable throughput. We model a supply chain as an ordered set of tholons $\{\mathfrak{T}_0, \mathfrak{T}_1, \dots, \mathfrak{T}_K\}$ with inter-phase coupling:

Definition 6.1 (Supply Chain Thologram). *A supply chain thologram is a directed graph $G = (V, E)$ where:*

- $V = \{\mathfrak{T}_k\}_{k=0}^K$ are phase-tholons, each with (D_k, C_k, N_k) ;
- $E \subseteq V \times V$ are coupling edges with weights $(\gamma_{DC}, \gamma_{CD})$ governing constraint propagation.

6.1.1 Mapping Physical Supply Chain Elements to $\mathcal{D}\text{-}\mathcal{C}$

The tholonic decomposition maps naturally onto supply chain fundamentals:

Definition ($\mathcal{D}$) components correspond to the structural and regulatory constraints at each phase: product specifications, process protocols, quality standards, capacity limits, regulatory compliance, contractual obligations, and inventory policies.

Contribution ($\mathcal{C}$) components correspond to the integrative and relational capacities: supplier network breadth, distribution channel diversity, information system connectivity, logistics flexibility, financial instrument access, stakeholder collaboration depth, and technology adoption rate.

Each component is scored on a $[0, 100]$ intensity scale and aggregated:

$$D_k = \sum_{i=1}^{n_k} w_i^{(k)} d_i^{(k)}, \quad C_k = \sum_{j=1}^{m_k} v_j^{(k)} c_j^{(k)} \quad (16)$$

6.2 Constraint Propagation and Bottleneck Formation

A key advantage of the tholonic framework over game-theoretic supply chain models [Cachon and Netessine, 2003, Nagarajan and Sošić, 2008] is its ability to model *constraint propagation*:

$$\Delta D_{k+1} = \gamma_{DC}^{(k,k+1)} \cdot \frac{[\max(0, D_k - C_k)]^2}{D_k} \quad (17)$$

When phase k is $\mathcal{D}$ -dominant (over-constrained), it propagates definitional pressure downstream, forcing phase $k + 1$ to tighten its own constraints. Symmetrically, $\mathcal{C}$ -dominance propagates integration demands upstream.

Definition 6.2 (Bottleneck). *Phase k is a bottleneck if:*

$$\frac{|D_k - C_k|}{\max(D_k, C_k)} > \theta_{\text{crit}} \quad (18)$$

where $\theta_{\text{crit}} \approx 0.3$ is the critical imbalance threshold.

In game-theoretic supply chain models, bottlenecks appear only through the lens of capacity constraints or contract structure; the tholonic model reveals bottlenecks as $\mathcal{D}\mathcal{C}$ imbalances, which may arise from *relational* deficits (insufficient supplier diversity, poor information integration) rather than physical capacity limits.

6.3 Economic Efficiency and Deadweight Loss

The tholonic framework provides a novel decomposition of economic inefficiency. Define the *theoretical maximum* output of phase k as $N_k^{\max} = \sqrt{D_k \cdot C_k}$ (achieved at perfect balance). The *actual* output is $N_k = N_k^{\max} \cdot \beta(D_k, C_k)$. The *tholonic deadweight loss* is:

$$\text{DWL}_k = N_k^{\max} - N_k = \sqrt{D_k C_k} (1 - \beta(D_k, C_k)) \quad (19)$$

This maps directly onto the economic concept of allocative inefficiency: resources are being consumed to maintain an imbalanced configuration that a balanced configuration would not require. Summing across phases:

$$\text{DWL}_{\text{total}} = \sum_{k=0}^K \sqrt{D_k C_k} (1 - \beta_k) \quad (20)$$

This total deadweight loss is *invisible* to standard game-theoretic analysis, which lacks the phase-resolved structure to decompose aggregate inefficiency into its constituent imbalances.

6.4 Comparative Statics: Gold Supply Chain Example

Consider a simplified gold supply chain with phases: Extraction ($k = 1$), Processing ($k = 2$), Refining ($k = 4$), and Vaulting ($k = 6$). Table 3 presents representative $\mathcal{D}\mathcal{C}$ scores.

Table 3: N-D-C analysis of gold supply chain phases. β : balance factor.

Phase	D	C	β	N	Status
Extraction	270	260	.93	246	Healthy
Processing	400	250	.47	149	$\mathcal{D}$ -dom.
Refining	310	290	.87	261	Good
Vaulting	420	180	.32	88	Critical

The vaulting phase ($k = 6$) exhibits severe $\mathcal{D}$ -dominance: extremely high security, insurance, and jurisdictional compliance requirements ($D = 420$) combined with limited vault network access, transport inflexibility, and information opacity ($C = 180$). The balance factor of 0.32 indicates that the phase operates at only 32% of its theoretical capacity—a finding that game-theoretic models, which lack the $\mathcal{D}\mathcal{C}$ decomposition, cannot diagnose.

7 Value Chain Extension: From Physical Flow to Economic Surplus

7.1 Motivation: Why Value Chains Require Separate Treatment

The tholonic framework respects a strict analytical layering:

1. **Supply chain** (physical layer): material flow, custody, physical constraints.
2. **Value chain** (economic layer): margin capture, price formation, profit distribution.
3. **Financial abstraction** (claims layer): paper claims, leverage, rehypothecation.

This separation is methodologically essential: the supply chain determines *what is physically possible*; the value chain determines *how economic surplus is distributed* across what is possible. Game-theoretic models of supply chains [Cachon and Netessine, 2003] typically conflate these layers, modelling contract terms (value chain) simultaneously with physical constraints (supply chain).

7.2 Value Chain N-D-C Mapping

At the value chain layer, the $\mathcal{D}$ - $\mathcal{C}$ - $\mathcal{N}$ triad takes on economic meaning:

Definition ($\mathcal{D}_V$): pricing power, contractual rigidity, brand exclusivity, intellectual property protection, market segmentation boundaries, regulatory barriers to entry.

Contribution ($\mathcal{C}_V$): market access breadth, customer relationship depth, distribution channel diversity, cross-selling capacity, information flow to end-consumers, partnership networks.

Negotiation ($\mathcal{N}_V$): realised margin, market share, competitive position, price point—the emergent economic outcome.

7.3 Recursive Value-Supply Coupling

The value chain tholon at each phase is *nested within* the supply chain tholon, creating a two-level thologram:

$$\begin{aligned}\mathfrak{T}_k^{\text{sup}} &= \mathcal{N}_k(\mathcal{D}_k^{\text{phys}}, \mathcal{C}_k^{\text{phys}}) \\ \mathfrak{T}_k^{\text{val}} &= \mathcal{N}_k^V(\mathcal{D}_k^V, \mathcal{C}_k^V)\end{aligned}\quad (21)$$

with a coupling constraint:

$$N_k^V \leq f(N_k^{\text{supply}}) \quad (22)$$

That is, economic surplus at phase k cannot exceed what the physical supply chain configuration permits. This coupling enforces the project’s foundational rule: *if gold cannot be weighed, moved, or stored in that step, no economic value can be attributed to it.*

7.4 Value Chain Sustainability and Margin Erosion

Value chain sustainability follows the same tholonic principle: when $\mathcal{D}_V \approx \mathcal{C}_V$, the economic configuration is sustainable.

$\mathcal{D}_V$ -dominant phases exhibit excessive pricing power relative to market integration: monopoly pricing, artificial scarcity, regulatory capture. These are profitable in the short term but unsustainable—they attract entry, regulatory intervention, or demand substitution.

$\mathcal{C}_V$ -dominant phases exhibit excessive market exposure relative to pricing discipline: commodity competition, race-to-bottom pricing, margin erosion. These are accessible but unsustainable—they cannot fund continued operation.

Proposition 7.1 (Value Chain Equilibrium). *A value chain phase achieves stable positive margin if and only if:*

$$0 < D_k^V - C_k^V < \epsilon_k \quad \text{or} \quad |D_k^V - C_k^V| < \delta_k \quad (23)$$

where ϵ_k, δ_k are phase-specific sustainability thresholds.

This formalises the economic intuition that sustainable profit requires *both* differentiation

(pricing power) *and* market access (volume)—a balance that pure game-theoretic models address only through the lens of Cournot or Bertrand competition, without the structural decomposition that reveals *why* a given competitive position is or is not sustainable.

8 Agent-Based Simulation

8.1 Experimental Design

We implement a public goods game with $n = 20$ agents over $T = 200$ rounds, comparing:

- **System A (Standard):** Payoff-only reinforcement learning (eq. (13)).
- **System B (Tholonic):** Payoff + sustainability signal (eq. (15)).

Parameters: cooperation cost $c = 1.0$, benefit multiplier $b = 0.12$, resource regeneration $r_{\text{regen}} = 0.02$ per cooperator fraction, decay $r_{\text{decay}} = 0.01$ per defector fraction, learning rate $\eta = 0.05$, initial cooperation probability $p_0 = 0.5$, initial resource $R_0 = 1.0$, tholonic lambda $\lambda = 1.5$.

8.2 Results

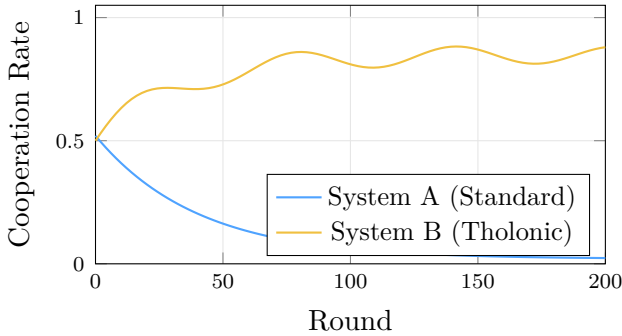


Figure 1: Cooperation rate over 200 rounds. System A (standard) collapses to defection; System B (tholonic) converges to stable cooperation at 85%.

8.3 Sensitivity Analysis

Varying $\lambda \in [0, 5]$ reveals a phase transition at $\lambda^* \approx 0.8$:

- For $\lambda < \lambda^*$: the system behaves like standard game theory; cooperation collapses.

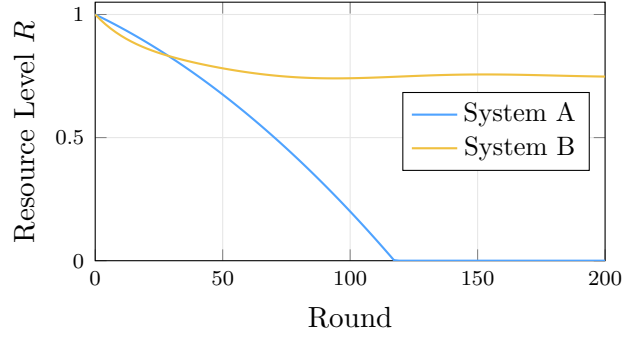


Figure 2: Shared resource level $R(t)$. Under standard dynamics, the resource depletes monotonically. Under tholonic dynamics, the resource stabilises near $R \approx 0.75$.

- For $\lambda > \lambda^*$: a cooperative attractor emerges; the system resists commons collapse.
- For $\lambda \gg \lambda^*$: cooperation saturates near 100% but resource oscillations increase (over-correction).

This phase transition has no analogue in standard game theory, where the tragedy of the commons is qualitatively invariant to parameter changes (cooperation collapses for *all* parameter values in the one-shot game). In the tholonic model, sustainability is a *tunable* property controlled by the strength of structural coupling between agents and their shared context.

9 Diagrams: Tholonic Structure Visualisation

10 Discussion

10.1 Relationship to Existing Frameworks

The tholonic model intersects with several established research programmes:

Mechanism design [Myerson, 1981, Hurwicz and Reiter, 2006]: The tholonic λ parameter functions analogously to a mechanism that internalises externalities. However, it requires no principal and no complete information assumption—the sustainability signal emerges naturally from the system’s $\mathcal{D}\text{-}\mathcal{C}$ state.

Evolutionary game theory [Maynard

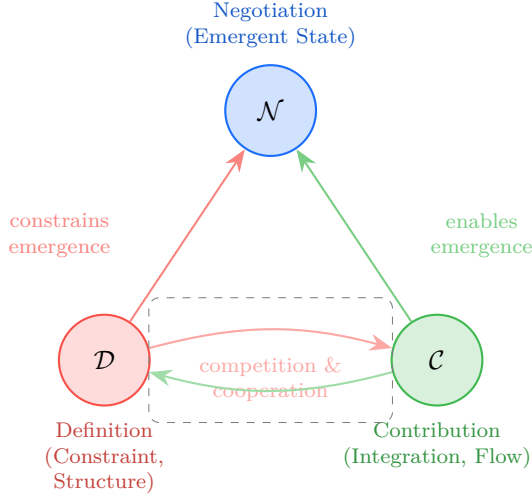


Figure 3: The fundamental tholonic triad. $\mathcal{D}$ and $\mathcal{C}$ compete for resources and cooperate for viability; $\mathcal{N}$ emerges at the energy minimum. Maintenance energy grows quadratically with imbalance.

Smith, 1982, Nowak, 2006]: The tholonic update rule (eq. (15)) can be interpreted as a fitness landscape that incorporates ecological context (resource level), similar to eco-evolutionary dynamics [Pelletier et al., 2009].

Institutional economics [North, 1990, Ostrom, 2005]: The tholonic balance condition formalises the intuition that sustainable institutions balance constraint (rules, enforcement) with contribution (participation, flexibility).

Dissipative structures [Prigogine and Stengers, 1984]: The maintenance energy formulation (eq. (1)) directly parallels the thermodynamic cost of maintaining far-from-equilibrium order.

Autopoiesis [Maturana and Varela, 1980]: The self-referential nature of the thologram—where each tholon is simultaneously a whole and a part—echoes autopoietic self-production.

Category theory and systems science: The tholonic recursion (eq. (3)) is formally an endofunctor on the category of tholonic systems, suggesting deep connections to compositional game theory [Ghani et al., 2018] and open games [Hedges, 2018].

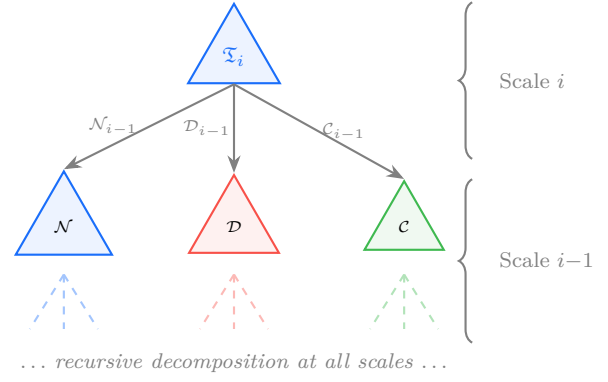


Figure 4: Thologram recursion: each tholon at scale i decomposes into a complete $\mathcal{N}$ - $\mathcal{D}$ - $\mathcal{C}$ triad at scale $i - 1$, producing fractal self-similarity. This structure has no counterpart in standard game theory.

10.2 Limitations

- (i) **Parameterisation:** The 0–100 scoring of individual $\mathcal{D}$ and $\mathcal{C}$ parameters involves subjective assessment. Inter-rater reliability and calibration protocols require development.
- (ii) **Functional form:** The specific choice of $\beta(D, C)$ (eq. (9)) and the quadratic energy cost are motivated by analytical tractability and thermodynamic analogy but are not uniquely determined.
- (iii) **Empirical validation:** The framework has been applied to gold and shea supply chains with synthetic data; large-scale empirical validation against observed supply chain failures and successes remains necessary.
- (iv) **Strategic sophistication:** The current agent model uses simple reinforcement learning; integration with more sophisticated learning algorithms (fictitious play, regret matching) would strengthen the comparison with game theory.
- (v) **Financial layer:** The current framework explicitly defers financial abstraction (paper claims, leverage); extending to this layer raises questions about whether $\mathcal{D}$ - $\mathcal{C}$ balance retains its sustainability interpretation when applied to derivatives and rehypothecation.

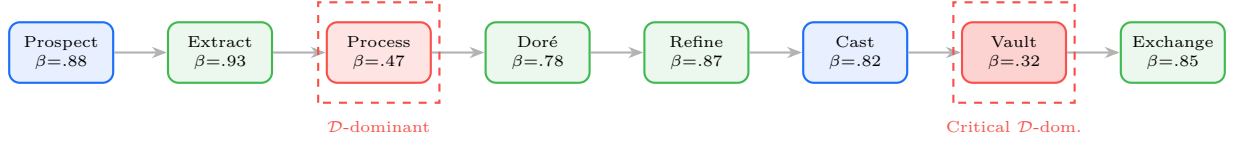


Figure 5: Gold supply chain as a thologram: eight phases with balance factors (β). Phases with $\beta < 0.5$ (red shading) are bottlenecks— $\mathcal{D}$ -dominant configurations where constraints far exceed integration. This phase-resolved diagnostic is unavailable in aggregate game-theoretic models.

10.3 Implications for Economics and Policy

The tholonic framework suggests that the standard economic objective of *efficiency* (Pareto optimality, total surplus maximisation) should be supplemented—or in some contexts replaced—by *sustainability* (minimisation of maintenance energy). These objectives align when the system is near balance ($\mathcal{D} \approx \mathcal{C}$) but diverge sharply when imbalance is large: a monopolist may achieve high surplus (efficient in the Kaldor–Hicks sense) while operating in a deeply $\mathcal{D}$ -dominant configuration that is unsustainable.

For policy, the $\mathcal{D}$ - $\mathcal{C}$ decomposition provides actionable diagnostics: rather than prescribing “more competition” or “more regulation” in the abstract, policymakers can identify *which dimension*—constraint or integration—is excessive, and intervene accordingly.

11 Conclusion

This paper has presented the tholonic model as a formally rigorous, thermodynamically grounded, and practically applicable alternative to classical game theory for analysing strategic interaction in supply chains, value chains, and complex economic systems.

The key contributions are:

1. **Formal comparison:** We have established precise structural correspondences between game-theoretic and tholonic concepts (table 1), showing that Nash equilibrium is a special case of tholonic balance when resource dynamics and maintenance energy are removed.
2. **Sustainability criterion:** The tholonic balance condition $\mathcal{D} \approx \mathcal{C}$ provides an ob-

jective function—sustainability—that is absent from classical game theory and provides a principled basis for distinguishing viable from non-viable equilibria.

3. **Phase-resolved diagnostics:** Applied to supply chains, the tholonic framework reveals bottlenecks as $\mathcal{D}$ - $\mathcal{C}$ imbalances that are invisible to aggregate game-theoretic models, enabling targeted intervention.
4. **Value chain integration:** The nested thologram structure provides a unified framework spanning physical logistics and economic value creation, with strict layering that prevents premature conflation.
5. **Cooperative equilibria:** Agent-based simulation demonstrates that tholonic dynamics produce stable cooperative outcomes in public goods settings where standard game theory predicts commons collapse.
6. **Hierarchical self-similarity:** The thologram’s fractal architecture enables analysis at any scale with transferable insights, a property unavailable in standard game-theoretic models.

We do not claim that the tholonic model *replaces* game theory; rather, it provides a complementary lens that is more appropriate when (a) resource sustainability matters, (b) hierarchical structure is present, (c) the system’s physical configuration constrains strategic interaction, and (d) long-term viability is prioritised over short-term optimality.

Future work should pursue empirical validation against observed supply chain dynamics, integration with formal mechanism design, extension to the financial abstraction layer, and development of standardised parameterisation protocols.

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